

Divyansh Shukla

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EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Expected 2027

CGPA: 9.03 / 10

EXPERIENCE

Power Pulse India

2026 – Present

Machine Learning Engineer Intern

India

- Trained YOLOv8n on the FLIR ADAS v2 thermal dataset to detect 6 road-safety classes: person, bicycle, car, motorcycle, bus, and truck.
- Ran multi-hour GPU training jobs on Modal with persistent volumes, stored secrets, and Weights & Biases logging, and added checkpoint resume so an interrupted run picks up where it stopped.
- Tuned training for single-channel thermal input: turned off hue and saturation augmentation, raised the classification loss weight to cut background confusion. Evaluation reports per-class AP50 and AP50-95 against a baseline of mAP50 0.59, mAP50-95 0.36.

CERTIFICATIONS

NPTEL: Applied Accelerated AI, Deep Learning with Computer Vision | **Coursera:** Deep Learning Specialization

TECHNICAL SKILLS

Languages: Python, C++, CUDA

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Ultralytics (YOLO), Stable-Baselines3, vLLM

Deep Learning: diffusion models, flow matching, transformers, attention mechanisms (RoPE, AdaLN-Zero), classifier-free guidance, object detection, reinforcement learning (PPO), vision-language models

LLMs & Agents: prompt engineering, structured output extraction, function calling, agentic workflows (planning, memory, tool use), retrieval-augmented search, LLM evaluation

AI Developer Tools: Claude Code, Gemini Antigravity, Gemini API

Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi, embedded/edge inference

Backend & Infrastructure: FastAPI, Django, Flask, Modal, Weights & Biases, Hydra, Git, Linux, NumPy, pandas

PROJECTS

RoPE-DiT: Diffusion Transformer with 2D RoPE

github.com/divyanshuklai/Custom2DRoPEDiT

- Wrote a ~5M-parameter Diffusion Transformer from scratch in PyTorch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Built a 2D rotary positional embedding that splits the attention head dimension into row and column halves and rotates queries and keys by each patch's height and width coordinates, so attention sees relative 2D distance instead of a flattened 1D index.
- Trained on CIFAR-10 with a rectified flow objective (MSE against the velocity field $v = x_1 - x_0$) and sampled with a hand-written Euler ODE solver and classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL

github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped with 30-dim proprioceptive observations, 12-dim joint-position actions, fractal terrain for domain randomization, and the Rapid Motor Adaptation bioenergetics reward with curriculum-scheduled penalty weights.
- Debugged a PPO baseline that ran 42M steps without learning a gait, tracing the exploding value loss to unnormalized observations, penalty dominance, and missing previous-action context.

Fashion Search: Conversational Product Discovery

github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app that answers natural-language product queries across 10 Indian e-commerce stores, using Tavily for search, Gemini 2.5 Flash for structured extraction, and BeautifulSoup for scraping.
- Wrote a priority-based image and price extractor that tries JSON-LD product schema, then Open Graph tags, then keyword-matched image tags. It is the baseline for an agentic version with planning, memory, and tool use.

ACHIEVEMENTS

Won the internal (intra-campus) round of the Smart India Hackathon at Vellore Institute of Technology, 2024.